

Assignment 1

Code Review and Error Correction

Name: Kanak Chaudhary

Internship: Python Programming Internship

Organization: VaultofCodes

Date: 04 July 2026

Code Snippet 1

Given Code

```
def add_numbers(a, b)
    return a + b
print(add_numbers(5, 10))
```

Error Identified

The colon (:) is missing after the function definition.

Corrected Code

```
def add_numbers(a, b):
    return a + b

print(add_numbers(5, 10))
```

Explanation

In Python, a colon (:) is required after a function definition. Adding it fixes the syntax error and allows the function to run correctly.

Code Snippet 2

Given Code

```
name = "Alice
print("Hello, " + name)
```

Error Identified

The string is missing the closing quotation mark.

Corrected Code

```
name = "Alice"  
print("Hello, " + name)
```

Explanation

Every string must begin and end with quotation marks.

Code Snippet 3

Given Code

```
for i in range(5)  
    print("Number:", i)
```

Error Identified

The colon (:) is missing after the for loop.

Corrected Code

```
for i in range(5):  
    print("Number:", i)
```

Explanation

A colon is required after loop statements in Python.

Code Snippet 4

Given Code

```
my_list = [1, 2, 3, 4, 5]  
print("The fifth element is: " + my_list[5])
```

Error Identified

The index is out of range, and a string cannot be concatenated with an integer.

Corrected Code

```
my_list = [1, 2, 3, 4, 5]  
print("The fifth element is:", my_list[4])
```

Explanation

Python lists start at index 0, so the fifth element is at index 4.

Code Snippet 5

Given Code

```
def greet(name):  
    print("Hello " + name)  
greet("Bob")
```

Error Identified

The print statement is not indented inside the function.

Corrected Code

```
def greet(name):  
    print("Hello " + name)  
  
greet("Bob")
```

Explanation

Indentation defines the function body in Python.

Code Snippet 6

Given Code

```
age = input("Enter your age: ")  
if age >= 18:  
    print("You are eligible to vote.")  
else:  
    print("You are not eligible to vote.")
```

Error Identified

The input value is a string and must be converted to an integer.

Corrected Code

```
age = int(input("Enter your age: "))  
if age >= 18:  
    print("You are eligible to vote.")  
else:  
    print("You are not eligible to vote.")
```

Explanation

The int() function converts the input into a number so it can be compared with 18.

Code Snippet 7

Given Code

```
def multiply(a, b):  
    result = a * b  
    return result  
print(multiply(4, 5))
```

Error Identified

The return statement is outside the function.

Corrected Code

```
def multiply(a, b):  
    result = a * b  
    return result  
  
print(multiply(4, 5))
```

Explanation

The return statement must be indented inside the function.

Code Snippet 8

Given Code

```
count = 10  
while count > 0  
    print(count)  
    count -= 1  
print("Countdown complete!")
```

Error Identified

The colon (:) is missing after the while statement.

Corrected Code

```
count = 10  
while count > 0:  
    print(count)  
    count -= 1  
print("Countdown complete!")
```

Explanation

Adding the colon correctly defines the while loop.